

SCI-NOI v0.1

SCI-NOI v0.1 r0.5 Scientific Rationale

Implementation-blind Stage B draft

Scientific owner: Grant Wilson

SCI-NOI_SCIENTIFIC_RATIONALE v0.1/draft-r0.5

Conditional owner-freeze candidate; not frozen. No implementation conformity, generator validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Status: implementation-blind proposed scientific-owner freeze candidate; scientifically freezeable conditional; not frozen.

Normative authority: the six ordered modules in SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.5, binding-file SHA-256 aa59ecaaaa149e2990d07623563d90af76c7b3084ee37c497a06e17ebf0fe213. This rationale explains that authority and authors no independent normative equation, requirement, assumption, definition, or prediction.

1. Scientific path

Exact retained PTC parent
v
Exact lossless balance representation -> assignment design
v
Active detector sign assignment -> frozen ordinary MAP operator
v
NOI-GEN members -> design-ignorable all-member admission
v
NOI-UNC conditional marginal second moment
v
NOI-STD independently realized MAP / conditional scale

Each arrow is an identity and admission boundary, not automatic realization. The sign enters once, only on the PTC signal occurrence. The unsigned denominator, projection, WCS, support, coverage, response, validity, and every other MAP gate remain fixed.

2. Scope, request state, and exact design

Base v0.1 owns one ensemble per exact observation and ToITEC array, with no cross-observation randomization or randomized coadd. not_requested and explicitly_disabled both have zero requested cardinality and no members or downstream products, but only the latter retains exact request/effective-plan identity plus disabling owner, policy, and cause. An enabled request is a positive integer and resolves atomically or fails completely.

Positive coefficient-mass detectors are grouped by exact readout network. Zero-mass, zero-total, ambiguous-membership, and singleton states keep their approved dispositions. Independent symmetric candidates are first-accepted under every exact network condition, with a positive finite cap, fail-closed exhaustion, replacement, valid duplicate draws, separately represented complement orbits, and no forced complement pair.

Every balance value is a canonical reduced arbitrary-precision rational. For each contribution, decoding the rational equals the exact contribution used by the frozen MAP parent. The encoding binds the parent and coefficient generations, exact projection and coefficient payload, representation source, and digest. It is never rounded from display text, re-estimated, independently quantized, or used to replace MAP numerical arithmetic. Without a lossless encoding, design resolution is unavailable.

3. Scientific target law and deterministic replay

For each active network stratum, the exact balance rule defines an admitted set A_h ; canonical stratum concatenation defines $A = \text{product}_h A_h$. Before exact key resolution, the declared scientific ideal model is independent uniform with-replacement draws from A . First-accepted rejection has this conditional law because every admitted candidate has the same base probability and the common earlier-rejection factor cancels.

The scientific randomization measure, the ideal candidate-bit/key-selection law, and the deterministic replay map are different objects. After key binding, the realized sequence is exactly $F(K, \text{plan})$. Replaying the same bound key, counters, serialization, and plan must reproduce the assignments independent of scheduling or layout. That exact equality establishes identity, not statistical independence, random-generator adequacy, or estimator uncertainty. Base v0.1 therefore adopts owner disposition B: iid uniformity is the declared ideal randomization model, while generator/randomness adequacy belongs to a separately named validation layer.

4. Protecting the target from member selection

The ideal Uniform(A) target survives member completion and admission only when their binding conjunction R_b is fixed before assignment or invariant under assignment/member output. Output amplitude, sign pattern, source imprint, map morphology, tail value, and assignment-dependent QC can be advisory, but cannot remove a member from the base estimator.

An assignment-dependent execution or numerical failure makes the exact base method unavailable. Regenerating the failed design under the same method identity until a favorable all-pass result appears is prohibited. If an output-dependent gate is scientifically necessary, it defines another method whose target is explicitly conditioned on selection and whose estimator, weights, uncertainty, and claim change accordingly. The unconditioned target cannot be retained by assertion.

5. Conditional UNC realization and STD meaning

Set/count equality is a realization condition, not a statement about every failed attempt. Realized initial UNC implies identical resolved/admitted member sets and equal positive requested, resolved, completed, and admitted counts. Only then does the known-zero-center estimator exist on the exact common domain with divisor N_{resolved} . If UNC is unavailable or failed, actual counts and causes remain factual and may be unequal; no survivor estimator or changed divisor exists.

The primary product remains `conditional_detector_sign_randomization_marginal_second_moment`. Its equality to target-law marginal variance follows from the known ideal target center, not from finite-ensemble centering. Source and structured residual imprint can remain. It is not physical-noise variance, total uncertainty, complete covariance, precision, or calibrated significance.

STD divides independently governed m_{MAP} by the compatible square-root conditional scale. It remains a nonlinear data-dependent product. MAP response divided by the scale is only a fixed-scale partial derivative; full response also needs separately authorized conditional-moment response and remains unavailable. The product has unit 1 and only the bounded standardization claim.

6. Route status

- Route/surface | Exact source/authority state | Gates before dependent realization or claim
- Parent and PTC/MAP/NOI boundaries | Identity source-closed by the exact admitted packet. | Exact MAP-facing coefficient/QC, lossless rational lift, numerical coverage_cut, MAP admission, support/WCS/response, and all frozen parent facts.
- Assignment design | Method/source closed; plan not automatically resolved. | Distinct request lifecycle; positive enabled cardinality; exact populations/arithmetic; exact tolerance/cap; complete key/counter plan; complete atomic resolution.
- Numerical GEN | Scientific method identified; numerical route unavailable here. | All parent/design gates, generation-input @2 owner approval and Registry/source binding, and successful realization. Changed use-profile authority is open.
- Initial UNC | Target, selection condition, and estimator identified; numerical product unavailable here. | Complete GEN; EQ-017 ignorability; member and ensemble @2 owner approval and Registry/source binding; all positive decisions; realized set/count implication; common domain; estimator uncertainty explicit or unavailable.
- Initial STD | Method and bounded claim identified; full response unavailable. | Independently governed MAP; compatible finite-positive moment; standardization @2 approval/binding; exact support/dependence/cause; separately authorized delta Vhat_cond for full response.
- Revised profiles | Four complete @2 records proposed; changed-use profile authority open. | Exact owner approval of bytes plus SCI-VAL source and Registry binding. Old or proposed names imply no action.
- Random-generator adequacy | Separate validation layer; unassessed. | Exact future validation method and evidence. Replay success is not a pass.
- Reciprocal and transformed routes | Reciprocal excluded; transformed numerical routes bounded but unavailable. | Separate owner authority, method/profile/use, exact external operator parity, and new generations as applicable.

Parent/boundary source closure does not close changed use-profile authority. Method identity, source closure, plan resolution, numerical realization, response-bearing realization, empirical calibration, implementation conformity, validation, and production authorization remain distinct states.

7. Proposed scientific-owner freeze disposition

The implementation-independent contract is coherent and scientifically freezeable conditional while exact numerical coefficient/coverage parents, ordinary numerical MAP/NOI realization, revised profile approval/binding, complete STD response, implementation conformity, and empirical validation remain unavailable or unassessed. Those conditions block their dependent products and claims; they do not prevent contract freeze. This package proposes that disposition but is not itself frozen.

No implementation conformity, random-generator validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, production suitability, or production authorization is claimed.